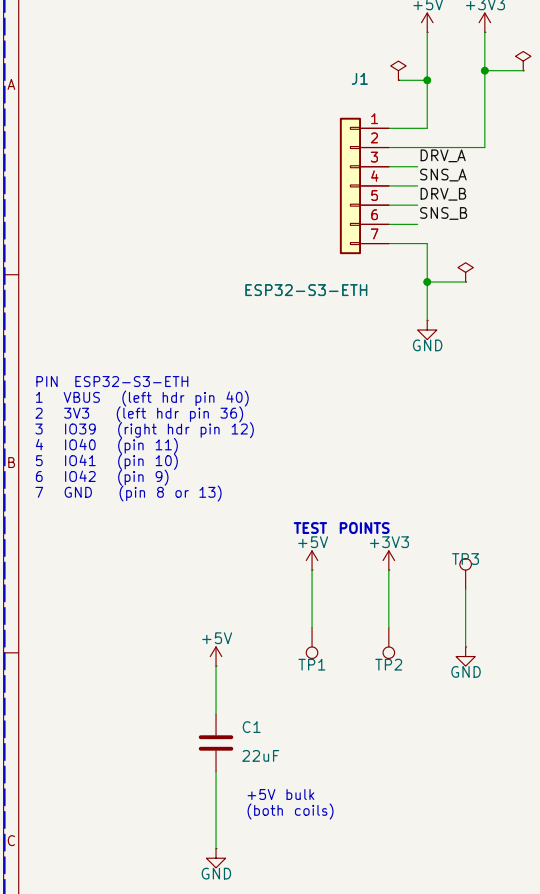
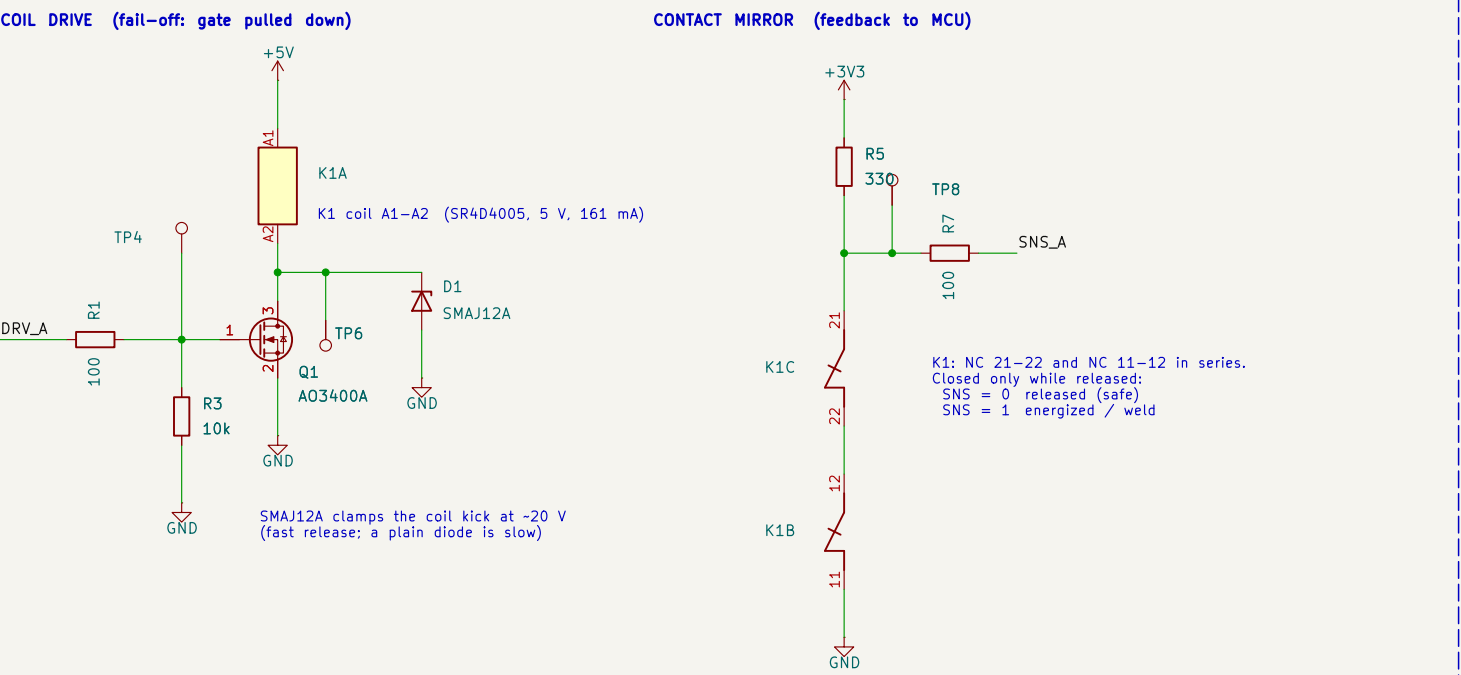


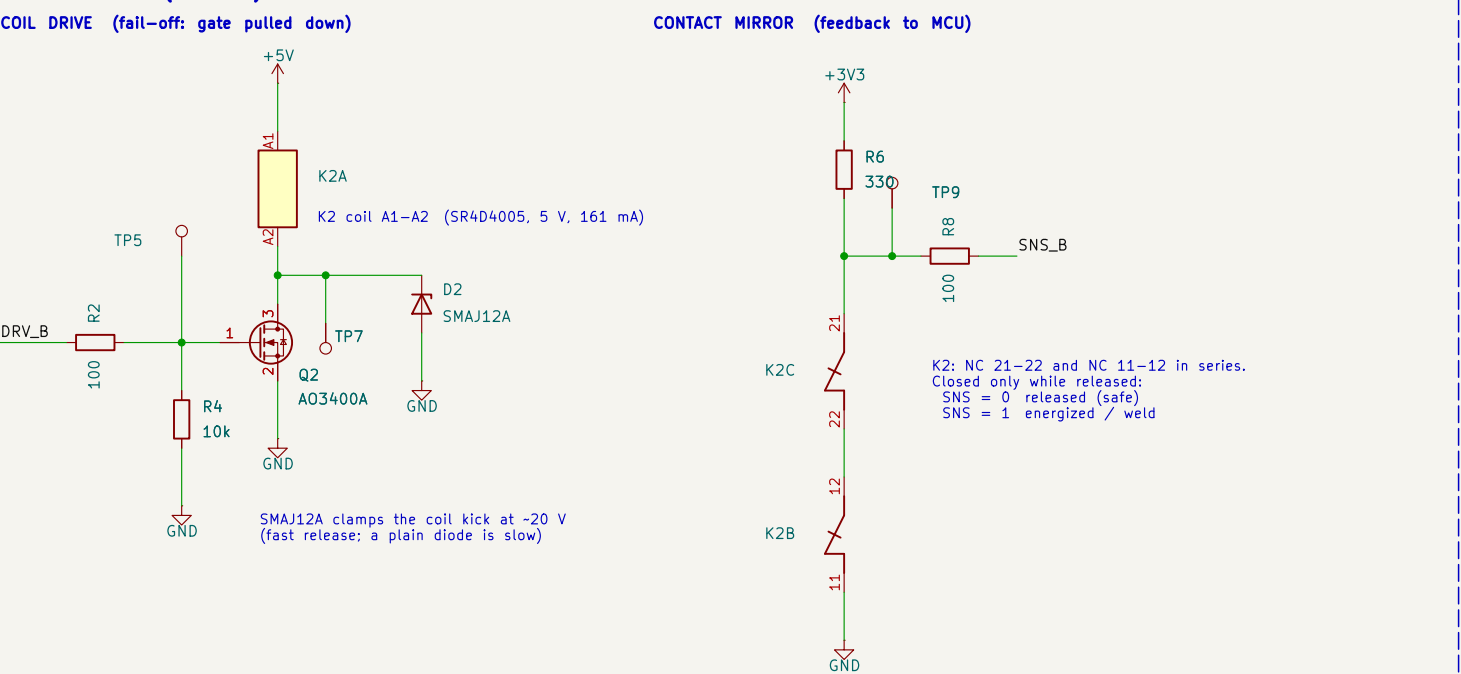
J1 ESP32-S3-ETH wire pads



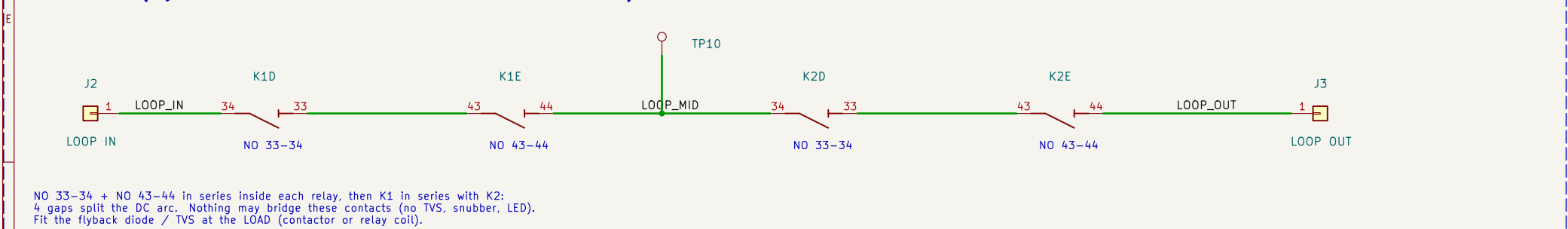
CHANNEL A (core 0): coil drive + contact mirror



CHANNEL B (core 1): coil drive + contact mirror



STOP LOOP (dry contacts, <= 24 V DC, <= 500 mA, inductive) - drawn released



HOW IT WORKS

Fail-safe: de-energized = loop open.
Power loss, reset, floating GPIO or a broken wire all leave both relays released.

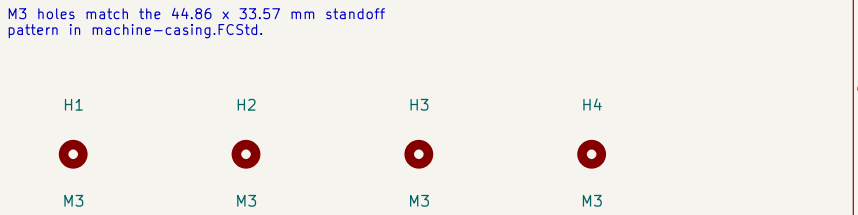
Each relay is force-guided (EN 61810-3 type A):
if a NO contact welds, every NC contact of that relay stays >= 0.5 mm open. The NC mirror therefore proves the loop contacts released.

Commanded 0, SNS reads 0: released, healthy.
Commanded 1, SNS reads 1: pulled in, healthy.
Commanded 0, SNS reads 1: WELD or coil stuck on.
The partner relay stops the robot; raise fault.
Commanded 1, SNS reads 0: did not pull in (safe).

Re-arm only after BOTH SNS read 0.

K1 = channel A = core 0 (IO39 drive, IO40 sense)
K2 = channel B = core 1 (IO41 drive, IO42 sense)
K unit letters: A coil, B/C NC mirror, D/E NO loop.

MECHANICAL



Schematic only. Not a functional-safety certification. See DESIGN.md		
NC contacts read back by the ESP32 (contact mirror) on IO40 / IO42		
Two TE SR4D4005 force-guided relays, NO contacts in series in the stop loop		
Polymath Robotics		
Sheet: /		
File: relay-board.kicad_sch		
Title: Machine relay board: dual force-guided safety relays		
Size: A3	Date: 2026-09-29	Rev: A
KiCad E.D.A. 10.0.0		Id: 1/1